

SDART: Spatial Dart AR Simulation with Hand-Trackled Input

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Figure 1: An example view of SDART: A user interacting with spatial darts.

Abstract

We present a physics-driven 3D dart-throwing interaction system for Apple Vision Pro (AVP), developed using Unity 6 engine and running in augmented reality (AR) mode on the device. The system utilizes the PolySpatial and Apple's ARKit software development kits (SDKs) to ensure hand input and tracking in order to intuitively spawn, grab, and throw virtual darts similar to real darts. The application benefits from physics simulations alongside the innovative no-controller input system of AVP to manipulate objects realistically in an unbounded spatial volume. By implementing spatial distance measurement, scoring logic, and recording user performance, this project enables user studies on quality of experience in interactive experiences. To evaluate the perceived quality and realism of the interaction, we conducted a subjective study with 10 participants using a structured questionnaire. The study measured various aspects of the user experience, including visual and spatial realism, control fidelity, depth perception, immersiveness, and enjoyment. Results indicate high mean opinion scores (MOS) across key dimensions.

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1 Introduction

Recent advancements in spatial computing [1–3] on head-mounted displays (HMDs) [4, 5] have enabled new forms of experiencing digital content, such as video games [6, 7]. High-quality stereoscopic displays provide realistic depth perception to the user in AR systems [8], and with the introduction of the intuitive input system of the AVP [9], which makes the user independent of any controller device, using an HMD has become seamless and natural. This input system utilizes multiple cameras and depth sensors for a full six degrees-of-freedom (6DoF) hand gesture detection. The 6DoF gesture [10] includes the 3 positional axes: *lateral*, *vertical*, and *depth*, and the 3 rotational axes: *pitch*, *yaw*, and *roll*, which enable the system to track the coordination and value of each component and apply mathematical formulas to simulate physics in an AR environment realistically.

Unity is a widely adopted development platform used for creating 2D, 3D, and XR applications [11] across operating systems, which promptly started supporting AVP development and has been

perfected at the one-year mark. The platform includes the PolySpatial framework to build spatial experiences that integrate directly with visionOS's bounded and unbounded AR volumes. In combination with Apple's ARKit [12, 13], where device-native features such as low-latency hand tracking [14], high-precision tracking, and scene understanding, the engine integrates input handling, physics simulation, and extra features into the application.

In this project, we aim to experiment with multiple cases of depth-sensitive games where spatial computing could potentially leverage the quality of experience [15, 16]. Therefore, as the first scenario, a real-world dart game is selected as it contains features that can benefit from the aforementioned technologies in an AR simulation [17]. As shown in Figure 1, the game involves players throwing pointed darts at a circular target (dartboard) fixed to a wall and requiring a standard distance. The board is divided into several scoring zones, visualized by numbers, and higher points for smaller or central zones such as the bullseye. This specific type of game includes the key aspects of accurate hand-eye coordination, precise control over angle and strength, and finding the right technique for each individual, which rewards both physical precision and spatial awareness - qualities that can be practically translated into AR interactions.

Our implementation recreates the core mechanics in a mixed reality environment, while preserving the enjoyment and immersiveness of the experience. The system spawns a dart pin on the player's hand with a pinch gesture and follows the 6DoF, allowing realistic physics-based motion, and simulates directional control, release timing, and throw force. Extra features include precise real-world distance measurements, a real-time scoring system, and two phases of training and main gameplay. The subjective results show that players have an enjoyable, realistic experience and performance in a spatial simulation compared to real physical darts, suggesting that spatial gaming could enhance immersiveness, a key aspect of interactive experiences.

2 System Design

The application is developed in three main modules, including (1) spatial rendering and object recognition, (2) spatial object tracking and physics simulation, and (3) object-to-object interaction and feedback. Using Unity's PolySpatial SDK, a virtual 3D dartboard is fixed in physical space, and the darts are spawned by detecting the player's hand as an object, including its position. In addition, a volume camera and scene anchor components are used to ensure the board maintains a stable position, and the dartpin motions are rendered realistically. Tracking hand movements and pinch gestures are performed via the ARKit SDK, where the dart stays attached to the hand and is thrown by detecting the pinch release. Calculation of the velocity upon release is done natively in Unity, while the spatial tracking information of position and rotation is the input of the ARKit. In case the dart pin collides with the dartboard, it sticks perpendicularly at the collision point to accurately detect and indicate the hit. Subsequently, it updates the score manager according to the nametag of the detected area on the dartboard, such as single, double, bullseye, etc. Finally, all user interface (UI) elements are also implemented as 3D objects for design consistency and immersiveness: the player's distance to the dartboard, and

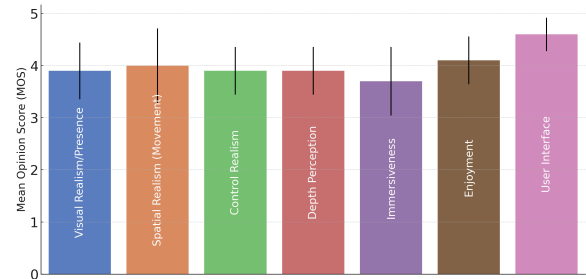


Figure 2: MOS for each questionnaire item.

scores are displayed on a separate board next to the dartboard, alongside play and reset buttons, which function both via pinch gesture and touch, identical to AVP's native input detection.

3 Experimental Study

We conducted a small-scale user study involving 10 participants, including 5 with prior experience in AR/VR gaming and 5 without. Each participant interacted with the system in two phases: (1) a tutorial mode, and (2) a main gameplay mode. In the tutorial mode, participants were allowed to practice freely and throw as many darts as they wished to become familiar with the interaction mechanics. In the main gameplay mode, they were limited to only three dart throws to achieve the highest possible score. Their virtual experience was mirrored on an external display for observation, and afterward, they completed a questionnaire consisting of seven 5-point Likert-scale items.

The questions include the main aspects of spatial gaming, including (1) **Visual Realism/Presence**: realism in the presence of the virtual objects in the real world, (2) **Spatial Realism (Movement)**: precise movement of the dart pins in the player's hand and in the space after throw, (3) **Control Realism**., naturalness and responsiveness of the hand-based input while interacting with darts, (4) **Depth Perception**: feeling the depth of space and object distances, (5) **Immersiveness**: immersiveness in playing virtual darts, (6) **Enjoyment**: player's enjoyment, and finally, (7) **User Interface**: the convenience provided by the user interface features. The results, as shown in Figure 2 indicate consistently high Mean Opinion Scores (MOS) across key aspects of the experience, including visual realism, control responsiveness, depth perception, and overall enjoyment. Participants particularly appreciated the natural hand-based interaction and intuitive user interface, which contributed to a strong sense of immersion.

4 Conclusion

The SDART system shows that physics-driven interactions and controller-free hand tracking on the Apple Vision Pro can recreate a realistic dart-throwing experience in AR. With our Unity implementation, users intuitively spawned, grasped, and threw virtual darts with realistic motion and scoring, creating a cohesive gameplay loop. A small user study—mixing experienced and novice AR/VR players—yielded high MOS for visual realism, control responsiveness, depth perception, immersiveness, and enjoyment, with narrow confidence intervals underscoring consistent agreement. Future work will extend this framework to multiplayer scenarios, adaptive difficulty, and more complex physics interactions to further explore spatial gaming on head-mounted displays.

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